

PROJECT TESTING PHASE

❖ 1.PURPOSE :

- This phase verifies that ComicCraft's AI pipeline, backend routes, and generated output work correctly and handle errors gracefully before deployment and documentation.

❖ 2.FUNCTIONAL TEST CASES :

TEST CASE	INPUT / ACTION	EXPECTED RESULT
VALID COMIC GENERATION	SUBMIT/GENERATE WITH A CLEAR PROMPT,CHARACTER,SETTING,TON E,STYLE.	OUTLINE,STORY,IMAGES,ANDPDF ARE GENERATE SUCCESSFULLY,COMIC PREVIEW PAGE LOADS.
MISSING FROM FIELD	SUBMIT/GENERATE WITH ONE REQUIRED.	FASTAPI RETURNS A VALIDATION ERROR INSTEAD OF CRASHING.
INVALID GEMINI OUTLINE RESPONSE	SIMULATE A MALFORMED/NON-JSON RESPONSE FROM GENERATE_OUTLINE.	VALUEERROR IS RAISED AND CAUGHT,ERROR INSTEAD OF APP CRASH.

IMAGE GENERATION FAILURE	TRIGGER/TEST-IMAGE WITH AN INVALID OR EMPTY PROMPT.	HTTPEXCEPTION (500) RETURNED WITH ERROR DETAIL, NOT A SILENT FAILURE.
JSON API GENERATION	SUBMIT/GENERATE-COMIC/JSON WITH A VALID JSON PAYLOAD.	RETURNS JSON WITH LAYOUT DATA AND PDF PATH.
PDF EXPORT	COMPLETE A FULL COMIC GENERATION.	PDF IS SAVED TO STSTATIC/EXPORTS AND PATH RETURNED CORRECTLY.
EXPORT SUCCESS FLOW	TRIGGER/EXPORT-SUCCESS WITH A VALID PDF_PATH	CONFIRMATION PAGE RENDERS WITH CORRECT PDF REFERENCE.

❖ 3. NON-FUNCTIONAL TEST :

- Performance: measure response time for outline, story, and image generation under normal prompts.
- Usability: confirm the form is understandable without prior technical knowledge.
- Consistency: verify tone, character, and art style remain coherent across all 5 panels in one comic.
- Error resilience: confirm try-except blocks in routes.py prevent server crashes on API failures.

❖ 4. LOCAL DEPLOYMENT VERIFICATION :

- Before functional testing, the server is started locally to confirm the environment is correctly configured:
- `uvicorn app.main:app—reload.`
- Homepage loads at `http://127.0.0.1:8000`.
- Interactive API docs load at `http://127.0.0.1:8000/docs`.
- Endpoints tested directly via `/docs`: `/generate`, `/generate-comic/json`, `/export-success`, `/test-image`.

❖ 5. SAMPLE TEST LOG :

Test Case	Input / Action	Expected Result
TC01	Prompt: "A brave fox exploring an enchanted forest"	5-panel comic generated, PDF downloadable– Pass
TC02	Empty story prompt submitted	Form validation error shown– Pass
TC03	<code>/test-image</code> with sample prompt	Image saved to static/panels, path returned– Pass

❖ CONCLUSION:

Testing confirms that ComicCraft's core pipeline (outline → story → images → PDF) functions correctly under normal and edge-case inputs, and that errors are handled without crashing the application. Verified functionality is carried forward into the Project Documentation phase.